CHILD WEARABLE NOTIFICATION SYSTEM

System Architecture Document

Phase 1 Deliverable

Date: June 2026 | Version: 1.0

1. System Overview

This document describes the system architecture for a child wearable notification bracelet, a device worn on a child's wrist that displays messages, images, and schedule reminders sent wirelessly by caregivers from any phone, tablet, or computer browser. The goal is a product that is simple to set up, reliable in daily use, and requires no technical knowledge from the staff.

The bracelet requires no interaction from the child. When a caregiver sends a message, the bracelet gently beeps and the image fills the screen automatically for a set duration before returning to the idle display showing the child's name and time. From the caregiver's side, a webpage accessible on any browser allows staff to select a child, send an image with an optional caption, and receive confirmation that the message was displayed.

Each bracelet connects to the facility's existing Wi-Fi network. All communication flows through a cloud relay service between the bracelets and the dashboard, meaning the caregiver does not need to be in the same room or building, the system works anywhere both sides have internet. The bracelet's location coverage matches wherever the facility Wi-Fi reaches, which in most indoor environments covers every room a phone would reach.

Battery life is engineered around how the bracelet actually operates. Two power strategies are offered — a **5-minute deep sleep** cycle for **maximum battery life,** and a **modem sleep mode** for **near-instant message delivery** and **faster out-of-range detection**. Both are covered in the power section with full analysis. Charging is via a sealed magnetic dock that requires no port opening, maintaining water resistance. When battery is low both the display and dashboard warn staff to collect the bracelet for charging.

The system monitors bracelet presence automatically. If a bracelet fails to check in, because a child moved out of Wi-Fi range, the bracelet was removed, or battery died, The dashboard raises an alert within minutes showing the child's name, last battery level, and time of last contact. Setup is a one-time process per bracelet requiring no technical knowledge from staff. The system supports up to **20 bracelets** simultaneously on a standard facility network.

Setup is a one-time process per bracelet. When powered on for the first time, the bracelet creates a temporary Wi-Fi network that a caregiver connects to using any phone or laptop. A setup page opens automatically in the browser where the caregiver enters the facility Wi-Fi details and gives the bracelet the child's name. If a bracelet ever needs to be reassigned or reconfigured, a recessed reset button on the enclosure, only accessible with a pen tip to prevent accidental activation by children, restores the device to setup mode.

In addition to sending live messages, the system supports a library of pre-loaded images stored directly on each bracelet, common schedule images such as meal time, bath time, or sleep time, which caregivers can trigger instantly from the dashboard without uploading anything. These images can be updated wirelessly at any time from the dashboard, keeping the library current without any physical access to the bracelets.

Question	Answer
Indoor range?	Full coverage anywhere the facility Wi-Fi reaches. Wherever a phone gets signal, the bracelet works.
Interference in crowded venues?	The bracelet uses the facility's existing Wi-Fi routers handle many devices. No special mitigation required beyond a functioning router.
Simultaneous bracelets?	Up to 20 on a standard facility router without infrastructure changes.

2. Hardware Architecture

All hardware is self-contained within the bracelet enclosure. No additional hardware is required at the facility beyond the existing Wi-Fi router.

Component	Part	Role
Microcontroller	ESP32-S3-WROOM-1 Module	Central processor. Runs all firmware, manages Wi-Fi, controls display, reads battery status, drives buzzer, manages sleep and wake cycles. Wi-Fi antenna integrated in module.
Display	ER-TFT1.69-3 (1.69" IPS, 240x280, ST7789V)	Shows messages, images, child name, time, battery status. 4-wire SPI. FPC ribbon connects directly to PCB via ZIF connector. 3.3V logic. 1000:1 contrast, 480 cd/m² brightness.
Battery Charger	BQ24074 (Texas Instruments)	Safe LiPo charging from USB-C input. Overvoltage and overcurrent protection. Charge status GPIO to MCU. 38µA quiescent.
Battery	400mAh Single-cell LiPo, 3.7V	Primary energy source.
LDO Regulator	AP2112K-3.3V	Converts battery voltage (3.2–4.2V) to stable 3.3V for MCU and display logic. 55µA quiescent, 250mV dropout.
Fuel Gauge	BQ27441-G1A (Texas Instruments)	Accurate battery percentage via I2C. Low-battery threshold detection. Replaces resistor divider — no continuous battery drain, no ADC inaccuracy. 90µA continuous.
Backlight Driver	AP3032 or equivalent LED driver	Regulated constant-current drive for display backlight LEDs. PWM brightness control from single ESP32 GPIO. Operates from battery voltage directly — no high-voltage boost required.
Buzzer	Passive piezo SMD	Two-tone alert on new message. GPIO via NPN transistor.
Charging Interface	Magnetic pogo-pin contacts (2-pin)	Sealed charging. No exposed port. Gold-plated spring pins on enclosure back mate with charging dock. Maintains water resistance.
Factory Reset	Recessed tactile switch	8-second long press clears provisioning data, reboots into setup mode. Tool-access only — child cannot activate accidentally.
Passives	MLCC capacitors, resistors, ESD diodes	Power rail stabilisation during Wi-Fi current spikes. ESD on charging interface. All ceramic — no electrolytics — moisture immune.

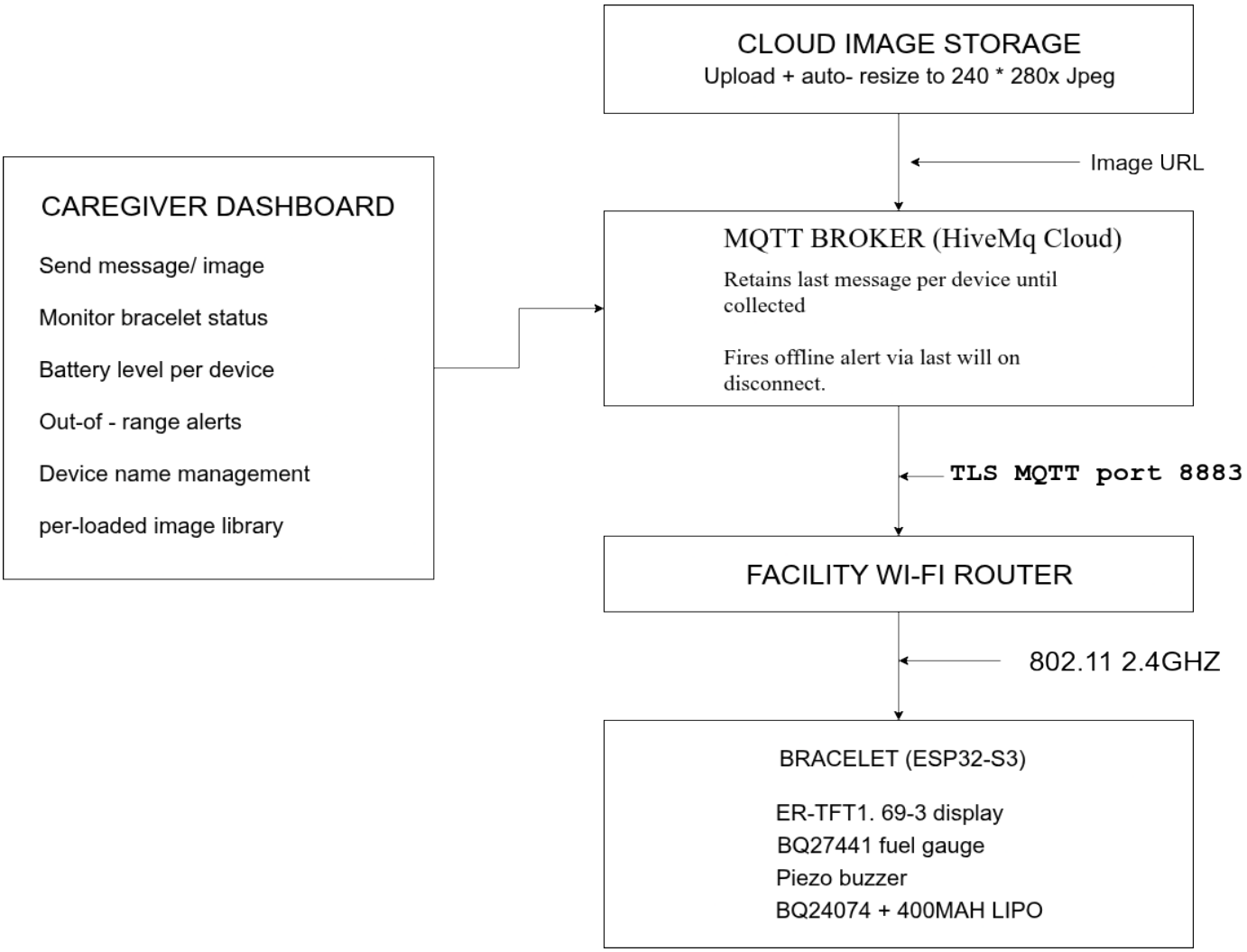
Physical notes: The ER-TFT1.69-3 panel footprint (approximately 35x42mm) gives sufficient area behind the display for the 400mAh LiPo pouch cell without the enclosure needing to widen beyond the display face. The antenna keepout zone on the PCB, copper-free, component-free, battery-free, is enforced during layout and is critical for reliable Wi-Fi range.

3. Communication, Software & Data Flow

3.1 Communication Stack

Layer	Technology	Purpose
Physical	Wi-Fi 802.11 b/g/n 2.4GHz	Bracelet to facility router
Security	TLS 1.2 on port 8883	Encrypts all data end-to-end
Messaging	MQTT v3.1.1	Lightweight protocol. Low overhead, guaranteed delivery, offline message retention at broker.
Image delivery	HTTPS GET	Bracelet fetches image from cloud storage by URL. MQTT carries the URL only — not the image binary.
Provisioning (one-time)	Wi-Fi SoftAP + captive portal	Bracelet creates temporary Wi-Fi network. Caregiver connects and enters credentials via browser. Works on any device including iPhone.

3.2 System Block Diagram



3.4 Message Journey — Caregiver to Child's Wrist

1. Caregiver selects a bracelet on the dashboard, uploads an image with optional caption, taps Send.
2. Dashboard auto-resizes image to 240x280px JPEG (~15–25KB) and uploads to cloud storage. A URL is generated.
3. Dashboard publishes MQTT message with image URL, caption, and message ID. QoS 1 (guaranteed delivery), retained flag set.
4. Bracelet wakes (deep sleep mode) or receives instantly (modem sleep). Connects to broker, receives message.
5. Firmware checks message ID against last-displayed ID in flash. If new, proceeds.
6. Bracelet fetches JPEG via HTTPS, decodes to RGB565 bitmap in RAM (~200ms), pushes to display over SPI (~100ms).
7. Buzzer produces two-tone alert. Child looks at wrist.
8. Bracelet publishes delivery acknowledgment with display timestamp. Dashboard confirms delivery.
9. Last message ID saved to flash. Bracelet returns to sleep (deep sleep mode) or stays connected (modem sleep).

3.5 Pre-Loaded Image Library

Common schedule images like meal time, bath time and sleep time are stored in bracelet flash memory during setup. Caregivers trigger them from the dashboard via a command containing only the image slot identifier, requiring no internet image download. This is faster, uses no data, saves battery and works even if internet is temporarily slow. The library can be updated wirelessly from the dashboard at any time without physical access to the bracelet.

3.6 Software Modules

Firmware (ESP32-S3)

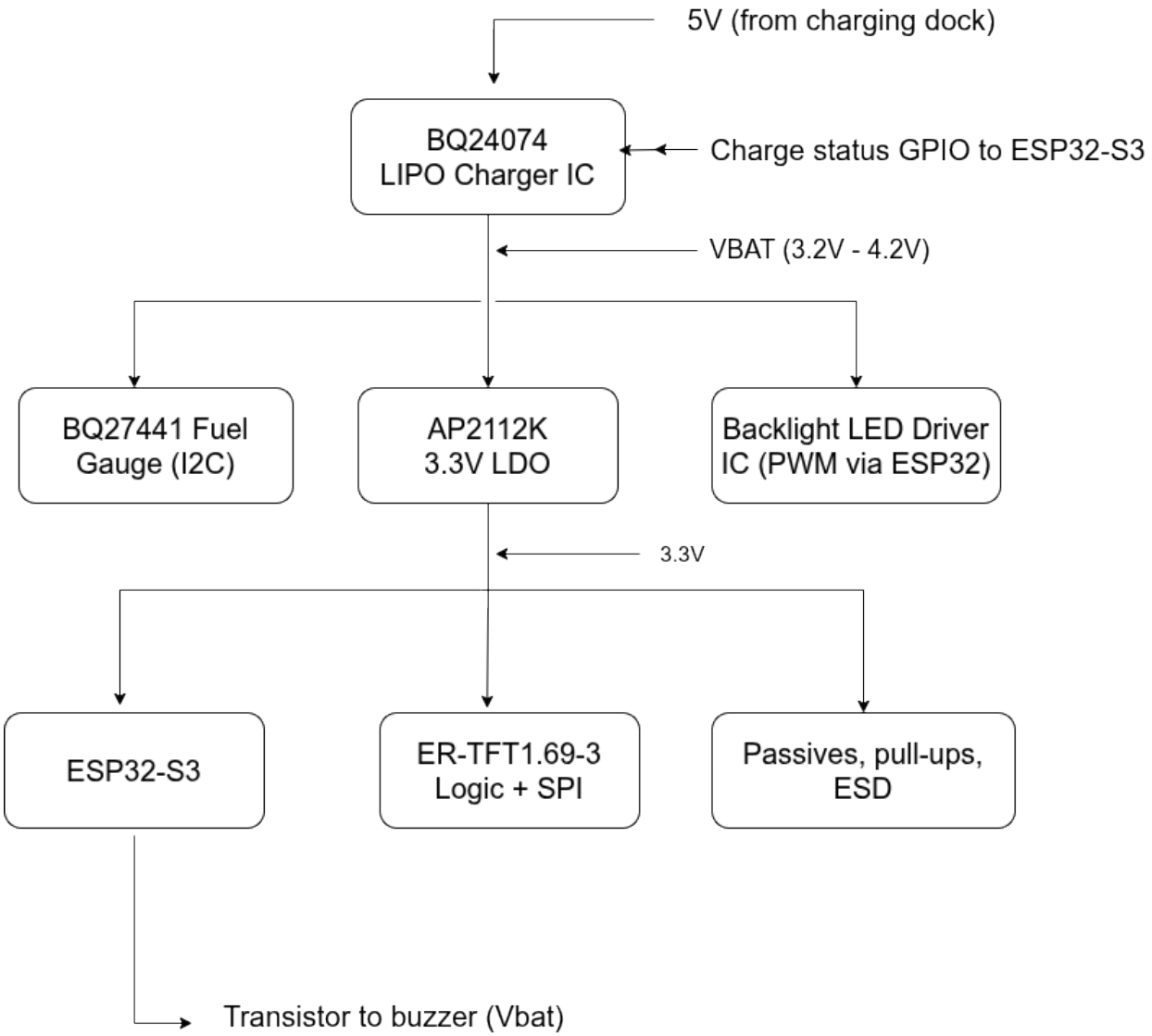
Module	Responsibility
Boot Manager	Checks NVS for provisioning flag. Routes to SoftAP setup or normal boot. Detects factory reset hold.
SoftAP Provisioning	Creates temporary Wi-Fi AP. Hosts captive portal. Receives and stores credentials and device name.
Wi-Fi / MQTT Manager	Connects to facility Wi-Fi and MQTT broker. Manages retries, TLS, Last Will, status publishing.
Wake / Sleep Manager	Controls deep sleep RTC timer (5-min mode) or modem sleep power mode. Single wake cycle handles both heartbeat and message check.
Image Handler	Downloads JPEG via HTTPS. Decodes to RGB565. Renders to display. Manages pre-loaded flash image slots.
Display Driver	Initialises ER-TFT1.69-3 via ST7789V SPI. Renders images, text, idle screen, icons. Controls backlight PWM. Sends SLEEP IN / SLEEP OUT commands.
Fuel Gauge Driver	Reads BQ27441 over I2C. Reports battery % to MQTT status payload.
NVS Manager	Stores Wi-Fi credentials, device name, last message ID, pre-loaded image data.

Backend & Dashboard

Service	Provider	Role
MQTT Broker	HiveMQ Cloud (free tier, 100 connections)	Message relay, retained messages, Last Will, QoS. Fully managed — no maintenance.
Image Storage	Cloudflare R2 or AWS	Stores and serves images. Auto-resizes on upload.
Dashboard	Netlify or Vercel	Hosts caregiver web app. Accessible from any browser

4. Power System Design

4.1 Power Flow



4.2 Component Selection Justification

Component	Part	Why Selected	Role
LiPo Charger	BQ24074 (TI)	38µA quiescent. Programmable charge current via single resistor. Built-in protection. Charge status GPIO. Proven	Safely charges 400mAh LiPo. Signals charge state to firmware.

Component	Part	Why Selected	Role
		application circuit.	
Fuel Gauge	BQ27441-G1A (TI)	Impedance Track algorithm gives accurate % not raw voltage. I2C interface. Low-battery threshold to MCU. 90MA, no continuous divider drain on battery. Eliminates ESP32 ADC non-linearity.	Accurate battery % to firmware and dashboard. Low-battery event without draining the cell.
LDO Regulator	AP2112K-3.3V	55µA quiescent. 250mV dropout — works down to 3.55V battery. 600mA output handles ESP32-S3 Wi-Fi peaks. SOT-25 compact package.	Stable 3.3V rail for MCU and display logic.
Backlight Driver	AP3032 or equivalent	Constant-current LED driver operating directly from VBAT. No high-voltage boost circuit needed. PWM brightness via single GPIO. Under 1mA quiescent.	Drives display backlight at regulated current. Enables firmware brightness control.
Battery	400mAh LiPo 3.7V	Power budget validated to provide 28–37 hours (deep sleep) or 12–20 hours (modem sleep) runtime. Physical dimensions fit within display panel size .	Primary energy source.
Decoupling	MLCC 100nF + 10µF ceramic	MLCC immune to moisture — no electrolytic failure risk. Values per ESP32-S3, BQ24074, and AP2112K datasheet application circuits.	Stabilise rails during Wi-Fi 500mA peak bursts.

All electrical values — quiescent currents, dropout voltages, capacitor values — are taken directly from manufacturer datasheets. No values are estimated.

4.3 Power Strategies

Strategy A — 5-Minute Deep Sleep

ESP32-S3 and Wi-Fi enter full deep sleep between cycles. Every 5 minutes the device wakes, connects to Wi-Fi and MQTT, checks for new messages (downloads and displays if present), publishes status, then returns to deep sleep. A single wake cycle handles both the status heartbeat and message check. The MQTT Last Will fires after one missed cycle plus a 5-minute grace period.

- Message delivery lag: up to 5 minutes
- Out-of-range detection lag: up to 5 minutes
- Battery life: best — see analysis below

Strategy B — Modem Sleep

ESP32-S3 stays awake. Wi-Fi radio dozes between router beacon intervals (~100ms) but the TCP and MQTT connection remains alive permanently. The MQTT stack sends keep-alive pings to the broker automatically with no firmware intervention needed, the processor handles this internally. Messages are received and displayed the moment the caregiver sends them. Out-of-range detection fires within 90 seconds of connection loss via the MQTT keep-alive timeout, with no firmware polling required.

- Message delivery lag: near instant (seconds)
- Out-of-range detection lag: under 2 minutes
- Battery life: lower, processor always awake

4.4 Component Current Reference (From Datasheets)

Component	Active / TX Current	Sleep / Idle Current	Source
ESP32-S3 + Wi-Fi TX	160–240mA (500mA peak)	0.02mA deep sleep	ESP32-S3 Datasheet, Table 10
ESP32-S3 in modem sleep	15–25mA average	N/A — processor always on	ESP32-S3 Datasheet, Section 3.7
ER-TFT1.69-3 logic (ST7789V)	5–7mA	0.03mA (SLEEP IN command)	ST7789V Datasheet, Section 6.1
Backlight at 35% PWM	17mA	0mA (driver off)	ER-TFT1.69-3 datasheet, LED spec
BQ27441 Fuel Gauge	—	0.09mA continuous	BQ27441 Datasheet, Section 7.5
AP2112K LDO	—	0.055mA quiescent	AP2112K Datasheet, Table 6.5
BQ24074 (not charging)	—	0.038mA quiescent	BQ24074 Datasheet, Section 6.3
Backlight driver IC	—	0.5mA quiescent	AP3032 Datasheet
Pull-up resistors (100kΩ)	—	0.033mA (3.3V/100kΩ)	Ohm's law
Piezo buzzer	15mA for ~1 second per alert	0mA	Buzzer component spec

4.5 Always-On Baseline Current

Component	Continuous Current
BQ27441 Fuel Gauge	0.090mA
AP2112K LDO quiescent	0.055mA
BQ24074 quiescent	0.038mA
Backlight driver IC quiescent	0.500mA
Pull-up resistors	0.033mA
Total always-on baseline	0.716mA

4.6 Power Budget — Strategy A: 5-Minute Deep Sleep

Per 5-minute cycle — no new message case:

Phase	Duration	Current	mAh per cycle
Deep sleep (baseline only)	~4 min 48 sec	0.716mA	0.057mAh
Wake + Wi-Fi connect +	~12 sec	200mA + 0.716mA	0.669mAh

Phase	Duration	Current	mAh per cycle
MQTT + status publish + no message		baseline	
Total per 5-min cycle (no message)			0.726mAh
Per hour (12 cycles)			8.71mAh/hr

New message addition — image download and display:

Phase	Duration	Current	mAh per event
Image download via HTTPS	~10 sec	200mA	0.556mAh
JPEG decode (processor only)	~0.2 sec	80mA	0.004mAh
Display image + backlight on	60 sec	22mA (logic + 35% backlight)	0.367mAh
Buzzer alert	~1 sec	15mA	0.004mAh
Total per new message event			0.931mAh

Overall battery life at 400mAh including new messages (20% derating applied for aging and temperature):

Message Frequency	Base mAh/hr	Message addition mAh/hr	Total mAh/hr	Theoretical runtime	Conservative runtime (20% derated)
0 messages/hr (heartbeat only)	8.71	0	8.71mAh	~46 hrs	~37 hrs
1 message per hour	8.71	0.93	9.64mAh	~41 hrs	~33 hrs
2 messages per hour	8.71	1.86	10.57mAh	~38 hrs	~30 hrs
1 message per 2 hours (0.5/hr)	8.71	0.47	9.18mAh	~44 hrs	~35 hrs

4.7 Power Budget — Strategy B: Modem Sleep

In modem sleep the processor is always awake at 15–25mA average. The MQTT stack sends keep-alive pings to the broker automatically — no firmware heartbeat needed. New message power is identical to Strategy A since both require the same Wi-Fi radio activity during image download.

Hourly power in modem sleep idle (no messages):

Component	Current	mAh/hr
ESP32-S3 modem sleep (processor on, radio dozing)	20mA average	20.0mAh
Always-on baseline (gauge, LDO, charger, pull-ups, driver)	0.716mA	0.716mAh
Total idle (no messages)		20.72mAh/hr

Overall battery life at 400mAh including new messages

Message Frequency	Idle mAh/hr	Message addition mAh/hr	Total mAh/hr	Theoretical runtime	Conservative runtime (20% derated)
0 messages/hr (idle only)	20.72	0	20.72mAh	~19 hrs	~15 hrs
1 message per hour	20.72	0.93	21.65mAh	~18.5 hrs	~15 hrs
2 messages per hour	20.72	1.86	22.58mAh	~17.7 hrs	~14 hrs
1 message per 2 hours (0.5/hr)	20.72	0.47	21.19mAh	~18.9 hrs	~15 hrs

4.8 Strategy Comparison Summary

	5-Min Deep Sleep	Modem Sleep
Message delivery lag	Up to 5 minutes	Near instant (seconds)
Out-of-range detection	Up to 5 minutes	Under 2 minutes
Battery life — 2 msgs/hr (conservative)	~30 hours	~14 hours
Battery life — 1 msg/hr	~33 hours	~15 hours
Battery life — 1 msg/2hrs (conservative)	~35 hours	~15 hours
Minimum requirement	10 hours	10 hours
Best suited for	Maximum battery life, tolerable message lag	Instant delivery, faster monitoring, daily charging acceptable

5. Range & Reliability

The bracelet connects to the facility router on 2.4GHz Wi-Fi — the same technology used by phones and laptops. In typical indoor environments with concrete walls and multiple floors, reliable coverage matches wherever a smartphone gets Wi-Fi signal. If dead zones exist, a standard Wi-Fi extender resolves them with no changes to the bracelet.

Out-of-range detection works at two levels. The MQTT broker fires a Last Will message automatically when a bracelet's connection drops unexpectedly. At the application level, a 3-minute grace period is applied before the dashboard raises a visible alert, preventing false alarms from momentary router hiccups.

Reliability Feature	How It Is Handled
Wake randomisation	Each bracelet's timer is offset by a random value at provisioning, preventing all devices hitting the router simultaneously.
MQTT QoS 1	All messages published with guaranteed delivery. Broker retries any unacknowledged message.
Retained messages	Broker holds the last message per bracelet until it connects. No messages lost during temporary disconnection.
Wi-Fi retry logic	Firmware retries connection up to 3 times before showing a no-connection indicator. Never requires manual restart.
Beyond 20 bracelets	Router capacity and broker tier should be reviewed before scaling beyond 20 devices.

6. Estimated Bill of Materials

Indicative pricing at prototype quantities (1–10 units) from Mouser and Digikey. Production quantities of 50+ will reduce per-unit cost.

Component	Part	Supplier	Est. Unit Cost (USD)
Micro-controller	ESP32-S3-WROOM-1	Mouser / Digikey	$3.00 – $4.50
Display	ER-TFT1.69-3 (FPC connector variant)	BuyDisplay (buydisplay.com)	$8.00 – $14.00
LiPo Charger	BQ24074RGTT	Mouser / Digikey	$1.80 – $2.50
Fuel Gauge	BQ27441DRZR-G1A	Mouser / Digikey	$2.50 – $3.50
LDO Regulator	AP2112K-3.3TRG1	LCSC / Mouser / Digikey	$0.30 – $0.60
Backlight LED Driver	AP3032 or equivalent	LCSC / Mouser	$0.30 – $0.80
Battery 400mAh LiPo	Custom sourced	Direct manufacturer	$2.50 – $4.00
USB-C Receptacle (power-only)	Generic power-only USB-C	LCSC / Mouser	$0.30 – $0.60
Piezo Buzzer	SMD passive buzzer	LCSC / Mouser	$0.20 – $0.50
Pogo Pin Contacts	Mill-Max or Harwin 2-pin	Mouser / Digikey	$1.00 – $2.00
Recessed Tactile Switch	TS-1187A or similar	LCSC / Mouser	$0.10 – $0.30
ZIF FPC Connector	Match ER-TFT1.69-3 FPC pin count and pitch	LCSC / Mouser	$0.50 – $1.00
MLCC Capacitors + Resistors + ESD	Various	LCSC / Mouser	$1.00 – $2.00
PCB Fabrication + Assembly	2-layer, JLCPCB PCBA	JLCPCB	$23.00 – $35.00

	Low	High
Total component cost per bracelet	$45.50	$72.30

7. Key Design Decisions & Trade-offs

Decision	Chosen	Rejected	Reason
Communication	Wi-Fi + MQTT	Bluetooth Low Energy	BLE requires caregiver's phone nearby at all times. Wi-Fi uses existing facility infrastructure and gives full-building coverage independently.
Image delivery	URL in MQTT, image via HTTPS	Full image binary in MQTT	MQTT is not designed for binary payloads. Separate HTTPS download keeps MQTT lightweight and enables the pre-loaded image library.
Battery monitoring	BQ27441 fuel gauge IC	Resistor divider on ESP32 ADC	Fuel gauge gives accurate %, handles threshold detection, and imposes no continuous drain. ESP32 ADC is non-linear; a divider draws current even during deep sleep.
Provisioning	SoftAP captive portal	BLE provisioning	SoftAP works on every device including iPhones without any app or browser restriction. Simpler for non-technical caregivers.
Display	ER-TFT1.69-3 IPS TFT (ST7789V)	Newhaven NHD-1.5-240240AF	NHD display requires up to 14V for its backlight boost circuit — a significant power design burden on a battery wearable. ER-TFT1.69-3 runs directly from battery via a simple LED driver. ST7789V has extensive ESP32 library support.
Charging	Magnetic pogo-pin dock	Exposed USB-C port	Pogo pins maintain full enclosure sealing for water resistance. No port opening means no moisture ingress path.
Power strategy	Two options offered: 5-min deep sleep or modem sleep	Single fixed strategy	Different deployments have different priorities. 5-min deep sleep maximises battery. Modem sleep maximises responsiveness. Both exceed the 10-hour requirement. Switchable via firmware update.
Parent interface	Web dashboard (browser)	Native mobile app	Web dashboard works on every device without installation. Native app requires separate iOS and Android development for no functional gain in this use case.

8. Items for Client Confirmation

The following assumptions were made during this architecture phase. Client confirmation is requested

Item	Assumption Made	Action Required
Facility Wi-Fi	Facility has 2.4GHz Wi-Fi with stable internet covering all areas children move in.	Confirm Wi-Fi exists and coverage is adequate.
Power strategy	5-minute deep sleep is the default. Modem sleep available as an alternative.	Confirm preferred strategy: maximum battery life (deep sleep) or near-instant delivery (modem sleep).
Message frequency	Average of 1–2 messages per bracelet per hour used for battery validation.	Confirm expected daily message frequency per child.
Message display duration	New messages display for 60 seconds before idle screen returns.	Confirm 60 seconds is appropriate for the children's age group.
Number of bracelets	Initial deployment fewer than 20 bracelets.	Confirm initial quantity and expected growth.
Image content	Still images only. No video or audio.	Confirm image-only is sufficient or advise if other media types are needed.
Pre-loaded images	A standard set of schedule images defined and pre-loaded at setup.	Provide list of standard images to be pe-loaded per bracelet.
Water resistance	IP67 (brief immersion) is the design target.	Confirm IP67 is required or advise if IP54 (splash only) is sufficient.
Alert grace period	5-minute grace period before out-of-range alert fires, to suppress false alarms.	Confirm 5 minutes is acceptable or advise preferred threshold.